



FOSSEE Semester Long Internship

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on

PCB Design using eSim

Submitted by

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CAN TO WIFI CONVERTER (DONGLE) FOR VEHICLE DIAGNOSTICS

1. INTRODUCTION

The CAN-to-Wi-Fi Converter Dongle is designed to provide wireless access to vehicle CAN communication data. It uses an ESP32-S3 as the main controller and Wi-Fi interface, along with a SN65HVD230 CAN transceiver for CAN communication. The PCB also integrates power supply, USB programming, status indication, and OBD connector sections. The complete design was developed using KiCad/eSim, covering schematic design, footprint assignment, PCB layout, and routing.

2. PROJECT OVERVIEW

The project focuses on designing a compact PCB that interfaces a vehicle's **OBD/CAN network** with a wireless Wi-Fi-based diagnostic system. The CAN transceiver receives CAN bus signals and communicates with the ESP32-S3, which processes the data and provides wireless connectivity. The design includes the necessary power, programming, protection, indication, and vehicle-interface circuitry required for the converter.

3. CIRCUIT DESCRIPTION

The CAN-to-Wi-Fi Converter Dongle is designed as a vehicle diagnostic interface consisting of a power supply section, ESP32-based MCU section, CAN communication interface, programmer/debugger interface, status indication LEDs, and an OBD connector.

Power Supply:

The power supply section accepts the vehicle battery input through the **V_BATT** line. It includes input protection, filtering, and voltage regulation circuitry to generate the required supply voltages for the different sections of the circuit.

MCU:

An ESP32-S3-WROOM-2 module is used as the main microcontroller and Wi-Fi interface. The MCU handles the communication between the CAN interface and the wireless diagnostic interface. The schematic also includes reset/enable and boot-related circuitry.

CAN Interface:

The CAN section provides the physical interface between the vehicle CAN network and the ESP32. It includes a SN65HVD230 CAN transceiver, termination circuitry, filtering/protection components, and CANH/CANL connections.

OBD Connector:

A 16-pin OBD connector is provided to interface the converter with the vehicle's diagnostic port. The connector carries the required CAN, battery, and ground connections.

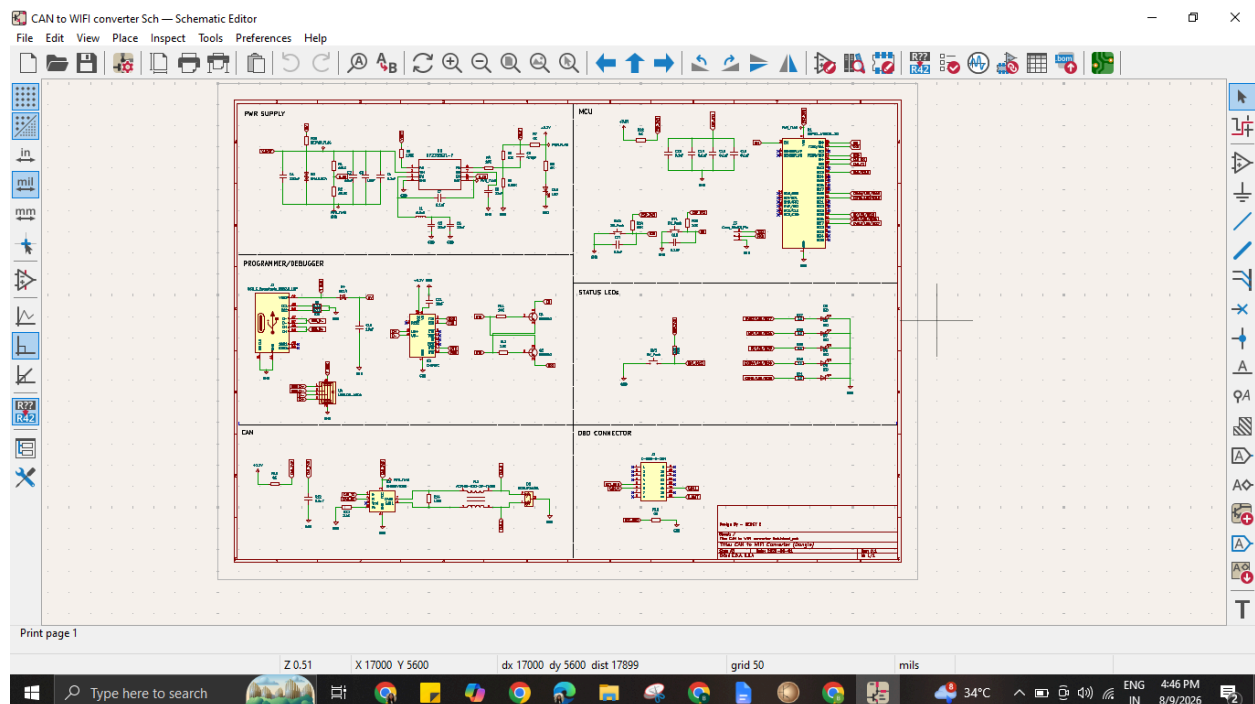
Programmer/Debugger:

A USB-C interface and CH340C USB-to-UART circuitry are provided for programming and debugging the ESP32. Additional circuitry is included for USB protection and automatic programming/reset control.

Status LEDs:

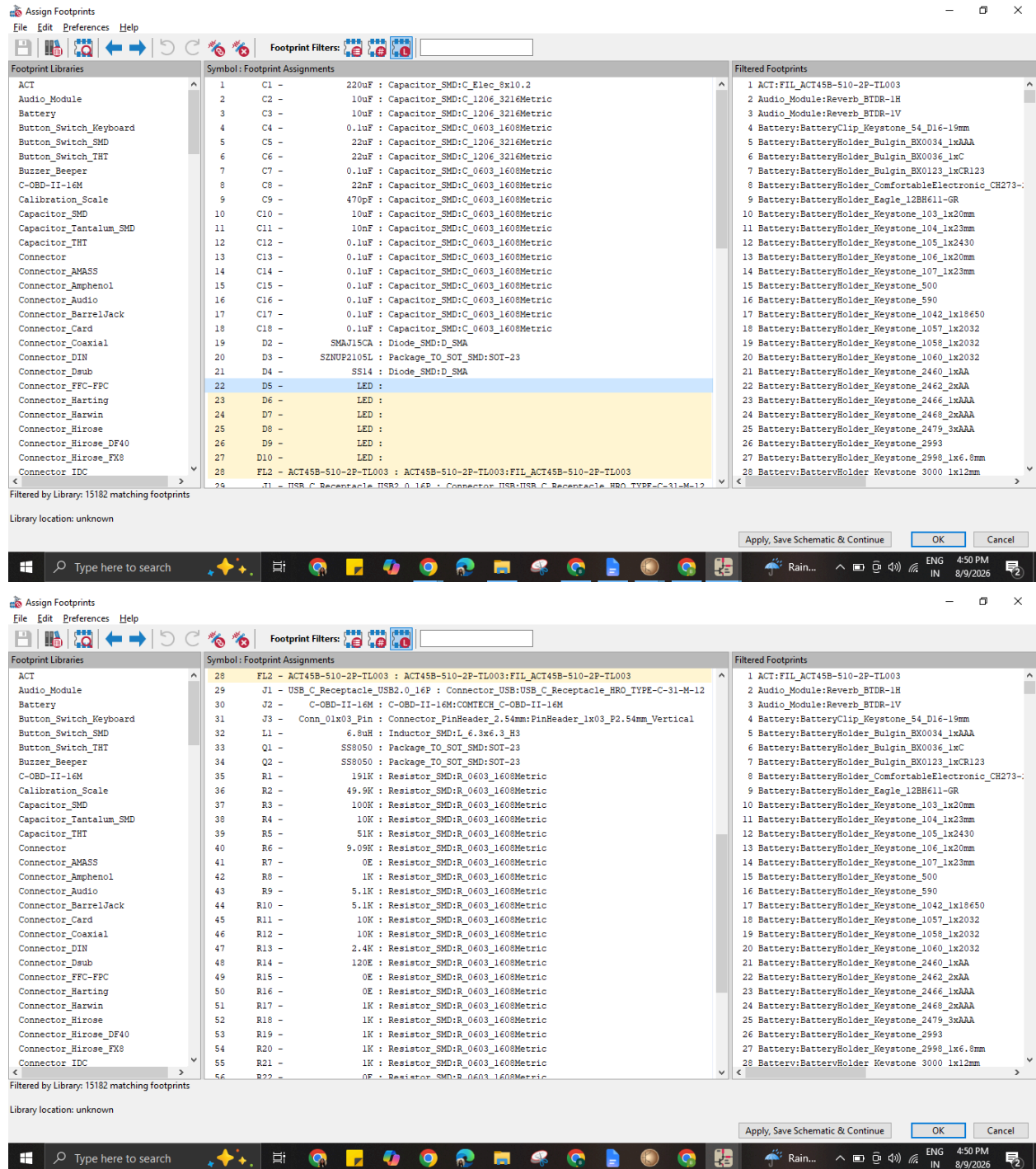
Multiple LEDs indicate different communication/status signals, including FAULT, Wi-Fi, Bluetooth, EXTRA, and COMM signals. This allows the operating status of the converter to be monitored visually.

4.SCHEMATICS

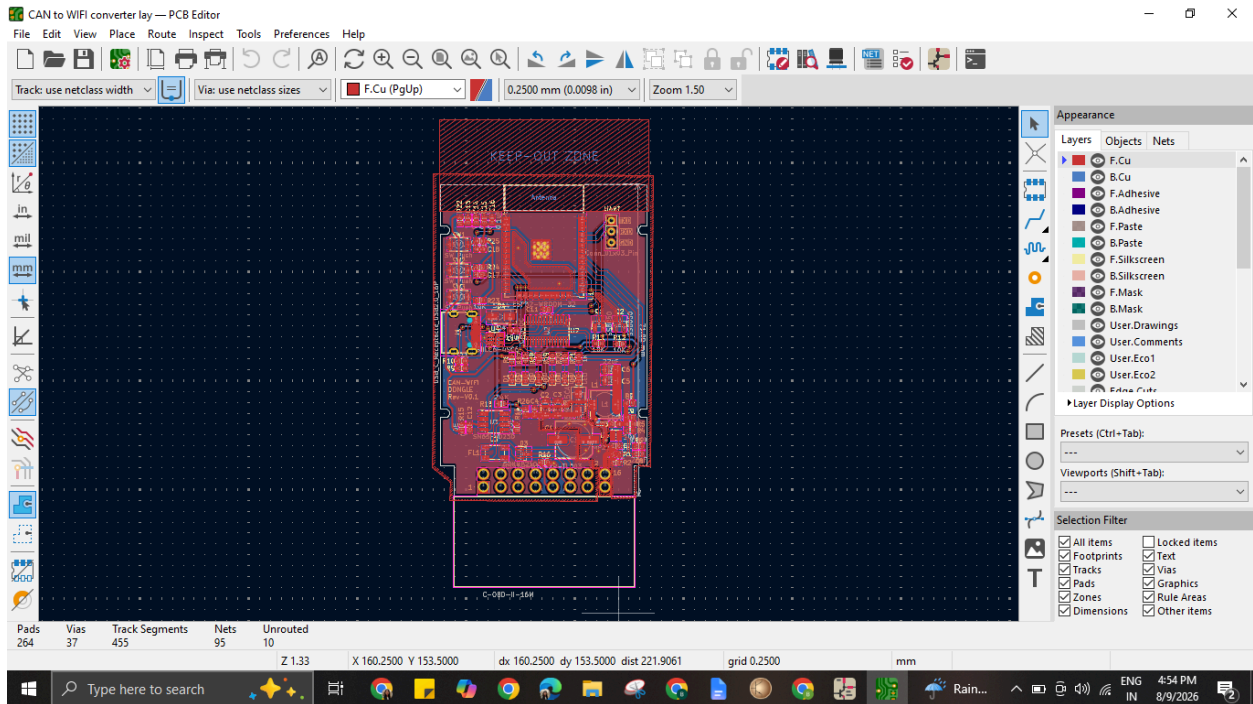


5.FOOTPRINT ASSIGNMENT

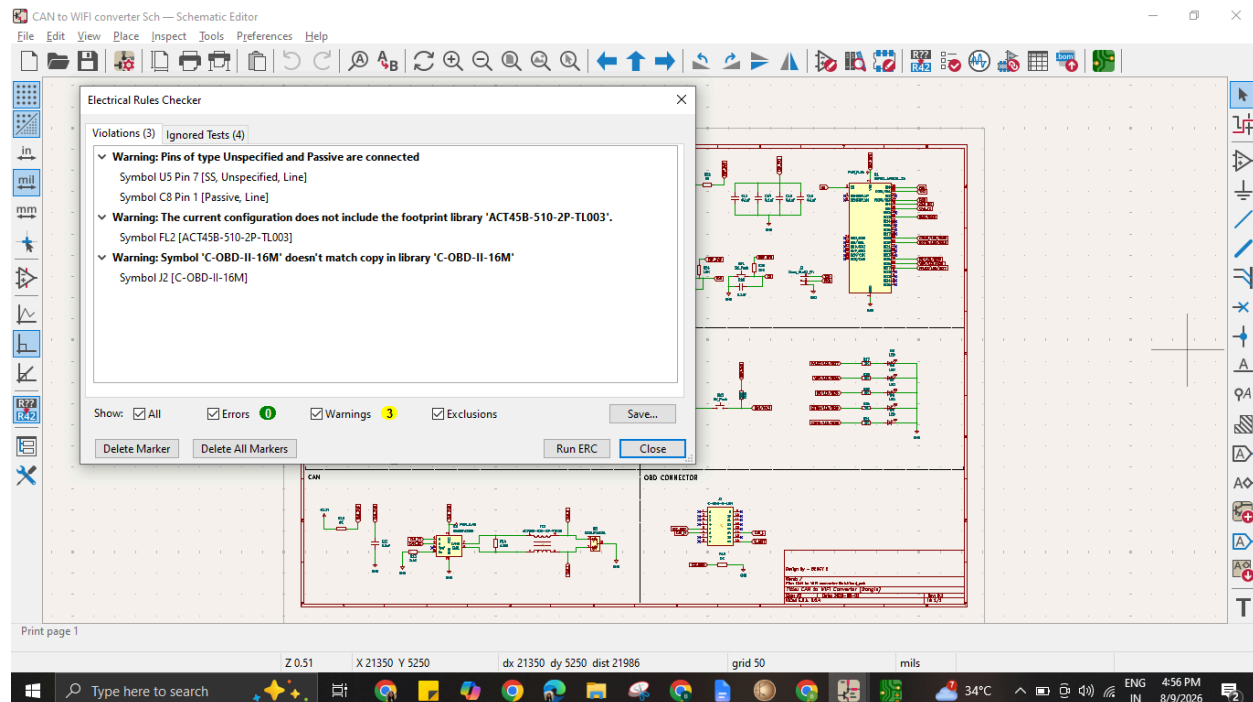
Appropriate PCB footprints were assigned to all schematic components based on their physical package and PCB mounting requirements.



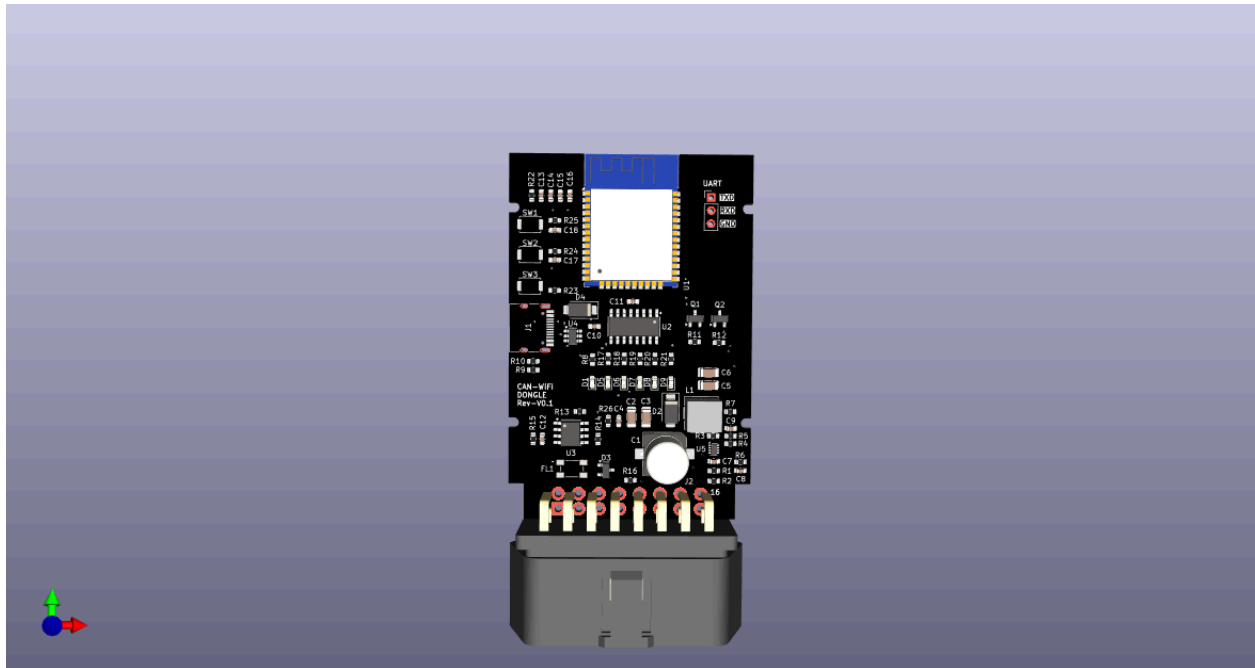
6.LAYOUT



7.DRC RESULTS



8.3D VIEW



9.CONCLUSION

The CAN-to-Wi-Fi Converter PCB was successfully designed with the required schematic, component footprints, PCB layout, and routing. The design provides a compact interface for vehicle CAN communication and Wi-Fi connectivity, with proper consideration of component placement, routing, and design rules.